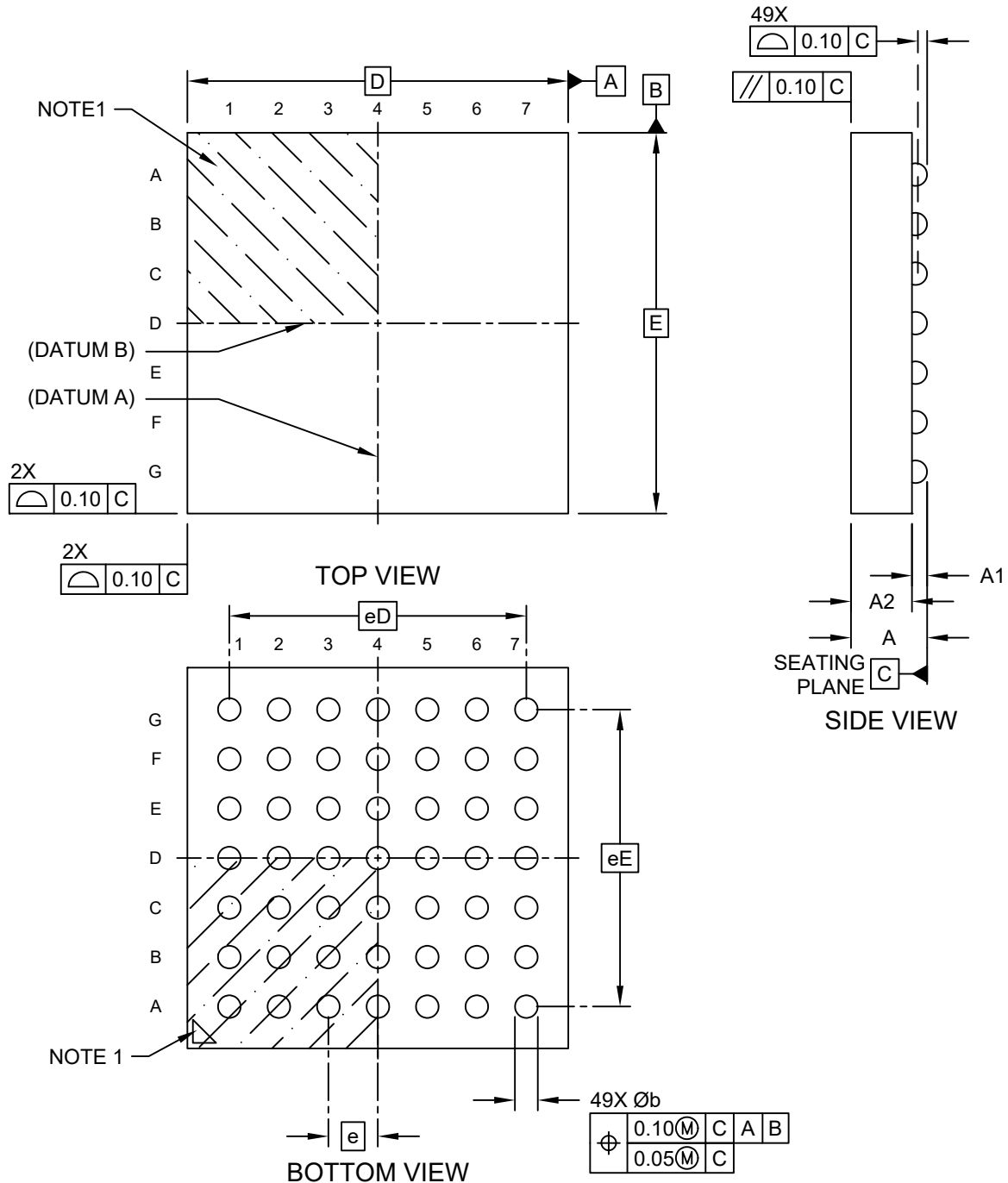


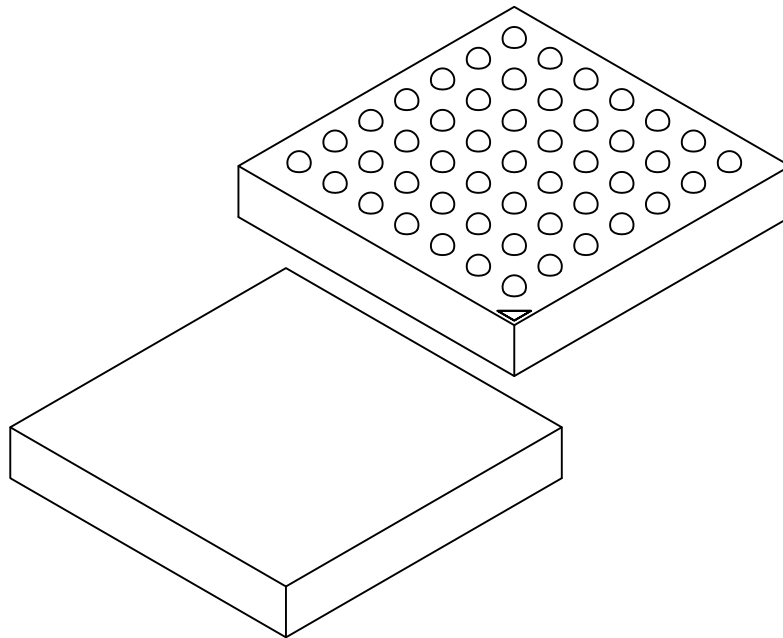
49-Ball Very Thin Fine Pitch Ball Grid Array (C7B) - 5x5x1.0 mm Body [VFBGA] Atmel Legacy Global Package Code CBD

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**49-Ball Very Thin Fine Pitch Ball Grid Array (C7B) - 5x5x1.0 mm Body [VFBGA]
Atmel Legacy Global Package Code CBD**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



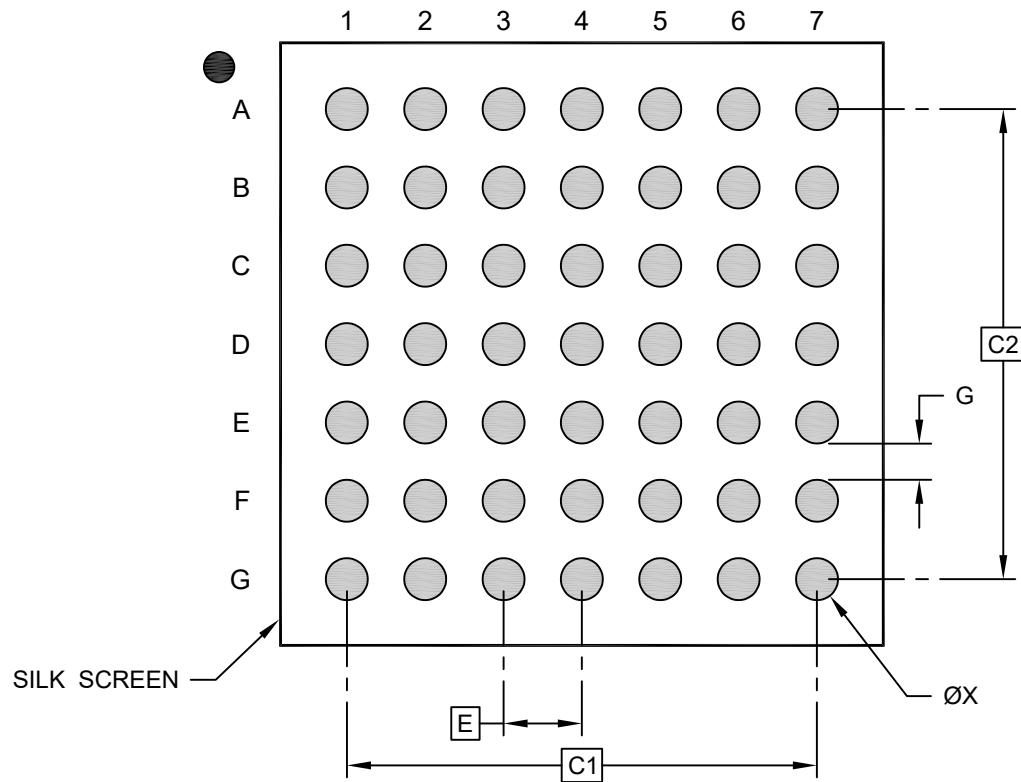
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	49		
Pitch	e	0.65 BSC		
Overall Terminal Spacing	eD	3.90 BSC		
Overall Terminal Spacing	eE	3.90 BSC		
Overall Height	A	–	–	1.00
Standoff	A1	0.20	–	–
Mold Cap Height	A2	0.58	–	–
Overall Length	D	5.00 BSC		
Overall Width	E	5.00 BSC		
Terminal Width	b	0.30	0.35	0.40

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

49-Ball Very Thin Fine Pitch Ball Grid Array (C7B) - 5x5x1.0 mm Body [VFBGA] Atmel Legacy Global Package Code CBD

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1	3.90 BSC		
Contact Pad Spacing	C2	3.90 BSC		
Contact Pad Diameter (X49)	X			0.35
Contact Pad to Contact Pad	G	0.30		

Notes:

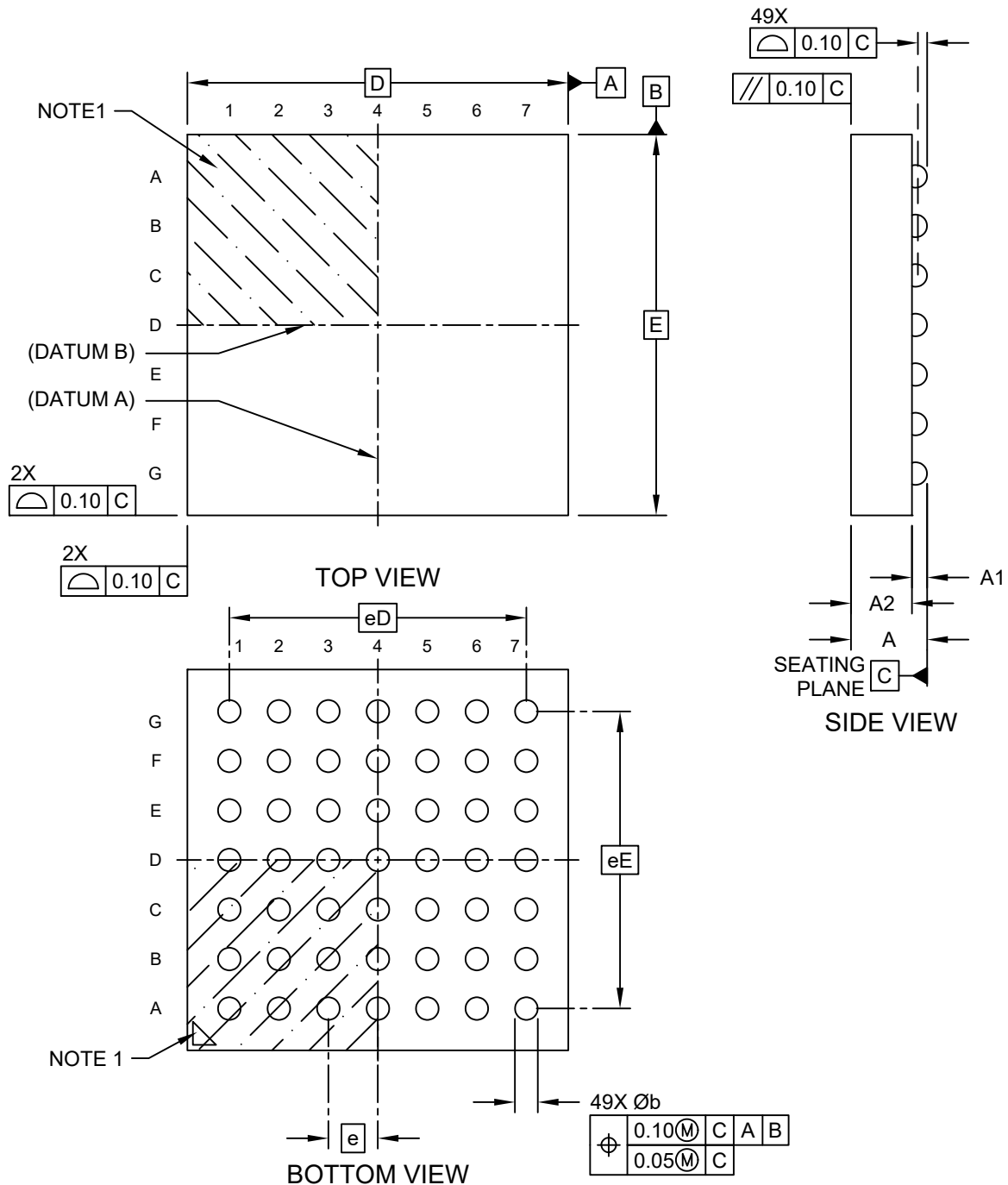
1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23168-C7B Rev B

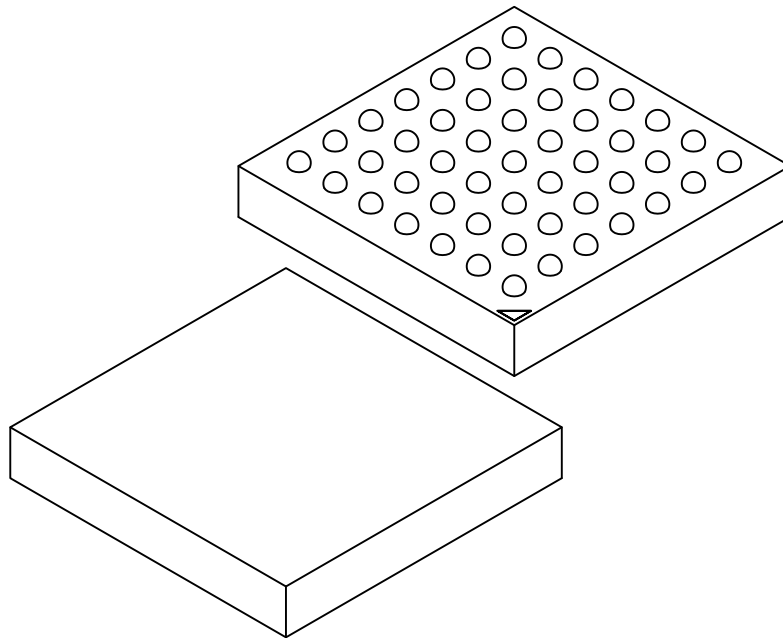
49-Ball Very Thin Fine Pitch Ball Grid Array (CTB) - 5x5x1.0 mm Body [VFBGA] Atmel Legacy Global Package Code CBD

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**49-Ball Very Thin Fine Pitch Ball Grid Array (CTB) - 5x5x1.0 mm Body [VFBGA]
Atmel Legacy Global Package Code CBD**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



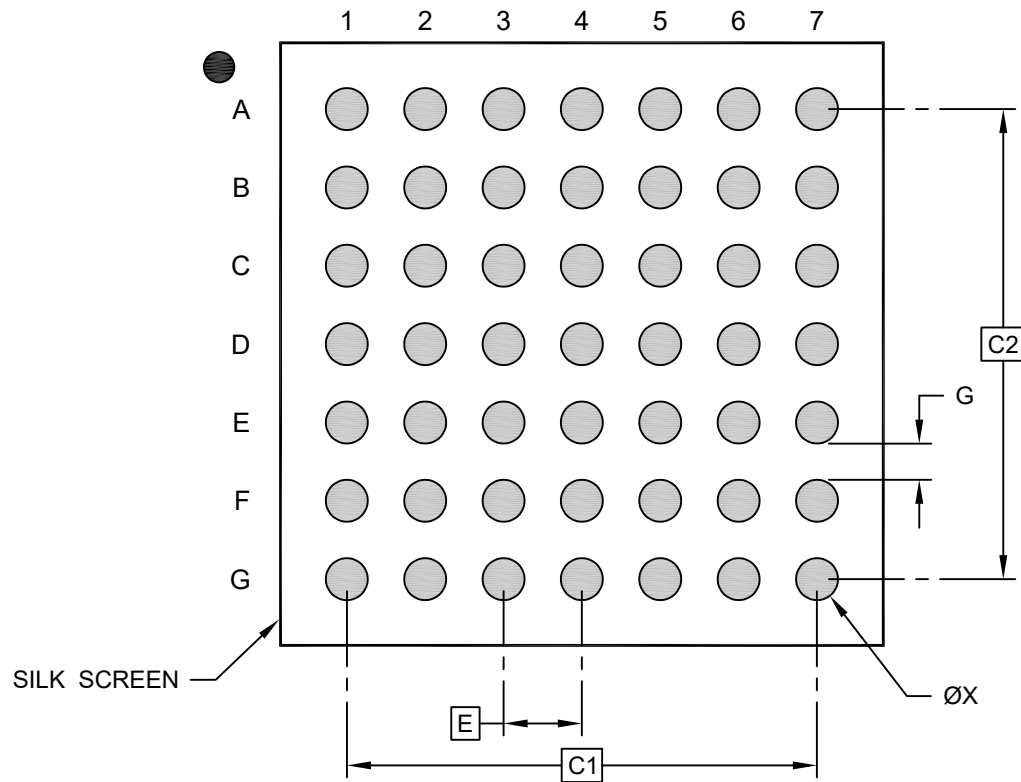
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	49		
Pitch	e	0.65 BSC		
Overall Terminal Spacing	eD	3.90 BSC		
Overall Terminal Spacing	eE	3.90 BSC		
Overall Height	A	–	–	1.00
Standoff	A1	0.20	–	–
Mold Cap Height	A2	0.58	–	–
Overall Length	D	5.00 BSC		
Overall Width	E	5.00 BSC		
Terminal Width	b	0.30	0.35	0.40

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

49-Ball Very Thin Fine Pitch Ball Grid Array (CTB) - 5x5x1.0 mm Body [VFBGA] Atmel Legacy Global Package Code CBD

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1	3.90 BSC		
Contact Pad Spacing	C2	3.90 BSC		
Contact Pad Diameter (X49)	X			0.35
Contact Pad to Contact Pad	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23168-CTB Rev B